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Assignment #2

Q1: $H_0: \mu = 35$ S.D = 1.8

$H_1: \mu != 35$

Var	N	Mean	StDev	SE Mean	Z	P
x	25	35.710	1.475	0.36	1.97	0.048

(a) $S.E = \frac{\sigma}{\sqrt{N}} = \frac{(1.8)}{\sqrt{5}} = 0.36$

$z = \frac{\bar{x} - \mu}{S.E} = \frac{35.710 - 35}{0.36} = 1.97$

P val at $\alpha = 0.05$: $p = 2[1 - \Phi(|z_0|)]$

$p = 2[1 - 0.976]$

$p = 0.048$

As $p < \alpha$ for $\alpha = 0.05$, we would reject the null hypothesis.

(b) Two sided test (:# sign. so checking both the above and below it)

(c) $\bar{x} - z_{\frac{\alpha}{2}} . S.E \leq \mu \leq \bar{x} + z_{\frac{\alpha}{2}} . S.E$

$35.71 - (1.96)(0.36) \leq \mu \leq 35.71 + (1.96)(0.36)$

$35.0044 \leq \mu \leq 36.4156$

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(d) $P = 1 - \phi(z_0)$

$P = 1 - 0.976 = 0.024$

Q2: $H_0: \mu = 12$

$H_1: \mu != 12$

Var	N	Mean	StDev	SE	T	P
x	10	12.564	0.936	0.296	1.91	$0.05 < P < 1$

(a) D.f = $n - 1 = 9$

(b) $StDev = (SE)(\sqrt{N})$

$S = (0.296)(\sqrt{10}) = 0.936$

$T = \frac{\bar{x} - \mu}{S.E} = \frac{12.564 - 12}{0.296}$

$T = 1.91$

~~$P = 2(\phi(|t_{.1}|)) - 2(\phi(|1.91|))$~~

~~$P = 2($~~

$P = 2[\phi(|t_{.1}|)] = 2[\phi(1.9)]$ ∴ Exact val not present

~~$1.833 < 2.262$~~ $0.05 \leq P \leq 0.1$

At $\alpha = 0.05$ we will not reject as $P > 0.05$

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(c) Two Sided Test (≠)

(d) $\bar{x} - \left(t_{(\frac{\alpha}{2}, df)}\right)(S.E) \leq \mu \leq \bar{x} + \left(t_{(\frac{\alpha}{2}, df)}\right)(S.E)$.

$12.564 - t_{(0.025, 9)}(0.296) \leq \mu \leq 12.564 + t_{(0.025, 9)}(0.296)$

$11.894 \leq \mu \leq 13.233$ → Do not reject at $\alpha = 0.05$

(e) For $H_1: \mu > 12$, it would be a one-sided test

$\mu \geq \bar{x} - t_{(\alpha, df)}\ S.E$

$\mu \geq 12.564 - (1.833)(0.296)$

$\mu \geq 12.02$

As μ does not lie in conf. interval, we will reject the null hypothesis

(f) For $H_0: \mu = 11.5$, $H_1: \mu \neq 11.5$

- Reject H_0 because μ does not lie in the interval from 2(d)

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Q3: $n = 17$, $S = 0.09s$

(a) $\alpha = 0.05$, $H_1: \sigma > 0.75$

$H_0: \sigma \leq 0.75$

$\chi^2 = \frac{(n-1)S^2}{\sigma^2}$, $df = 16$

$\chi^2 = \frac{(16)(0.09)^2}{(0.75)^2} = 0.2304$

$P > 0.995$

As $P > \alpha$, do not reject H_0

(b) $\alpha = 0.05$

$$\sqrt{\sigma^2} \geq \sqrt{\frac{(n-1)S^2}{\chi^2_{\alpha,(n-1)}}}$$

$$\chi^2_{0.05,16}$$

$$\sigma \geq \sqrt{\frac{(16)(0.09)^2}{26.30}}$$

$\sigma \geq 0.0702$

As $0.75 \geq 0.0702$, do not reject H_0

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Q4: $N = 200$ $\quad \hat{p} = \frac{16}{200} = 0.08$

$H_0: P \geq 0.10$

$H_1: P < 0.10$ $\quad q = 0.92$

(a) $z = \frac{\hat{P} - P_0}{\sqrt{P_0 q_0 / n}} = \frac{0.08 - 0.10}{\sqrt{(0.10)(0.9)/200}} = -0.943$, $\alpha = 0.01$

$P_v = \Phi(z_0) = \Phi(-0.94)$

$P_v = 0.17360$

As $P_v > \alpha$, we do not reject H_0

(b) $P \leq \hat{p} + z_\alpha \sqrt{\frac{\hat{p}\hat{q}}{n}}$

$P \leq 0.08 + (2.33)\sqrt{\frac{(0.08)(0.92)}{200}}$

$P \leq 0.125$

As $0.10 < 0.125$, do not reject H_0.

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Q5 $\sigma_1 = \sigma_2 = 3$, $n_1 = 20$, $n_2 = 20$

$\bar{x}_1 = 18$ $\bar{x}_2 = 24$

(a) $H_0: \mu_1 = \mu_2$ $\alpha = 0.05$

$H_1: \mu_1 \neq \mu_2$

$$z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{(18-24) - (0)}{\sqrt{\frac{3^2}{20} + \frac{3^2}{20}}} = -6.325$$

$$P = 2[1 - \Phi(|-6.325|)] = 2[1 - \Phi(6.33)]$$

$$P = 2[1 - 0.999999] = 0.000001$$

(b) $$(\bar{x}_1 - \bar{x}_2) - Z_{\frac{\alpha}{2}} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \leq \mu_1 - \mu_2 \leq (\bar{x}_1 - \bar{x}_2) + Z_{\frac{\alpha}{2}} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

$$(18-24) - (1.96)\sqrt{\frac{3^2}{20} + \frac{3^2}{20}} \leq \mu_1 - \mu_2 \leq (18-24) + (1.96)\sqrt{\frac{3^2}{20} + \frac{3^2}{20}}$$

$$-7.86 \leq \mu_1 - \mu_2 \leq -4.14$$

As $\mu_1 - \mu_2 = 0$ is not in the conf interval, we reject H_0

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Q6: $H_0: \mu_1 = \mu_2$ $n_1 = 15$ $n_2 = 15$ $\alpha = 0.05$

$H_1: \mu_1 \neq \mu_2$ $\bar{x}_1 = 4.7$ $\bar{x}_2 = 7.8$ $\sigma_1 = \sigma_2$

$s_1^2 = 4$ $s_2^2 = 6.25$

(a) $$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{(4.7 - 7.8) - 0}{2.26\sqrt{\frac{1}{15} + \frac{1}{15}}} = -3.76$$

$$S_p = \sqrt{\frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}} = \sqrt{\frac{(15-1)(4) + (15-1)(6.25)}{15+15-2}}$$

$S_p = \approx 2.26$ $d.f = 28$

$P = 2[\phi(|-3.76|)] = 2[\phi(3.76)]$

$P \approx 2(0.0005)$

$P \approx 0.001$ ∴ Table Limitation → $P < 0.05$, reject

(b) $$(\bar{x}_1 - \bar{x}_2) \pm t_{(\frac{\alpha}{2}, n_1+n_2-2)}\sqrt{S_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$

$$(4.7 - 7.8) \pm \overbrace{t_{(0.025,28)}}^{2.048}\sqrt{(2.26)^2\left(\frac{1}{15} + \frac{1}{15}\right)} \leq \mu_1 - \mu_2 \leq (4.7 - 7.8) + t_{(0.025,28)}\sqrt{(2.26)^2\left(\frac{1}{15} + \frac{1}{15}\right)}$$

$$-4.79 \leq \mu_1 - \mu_2 \leq -1.41$$

As $\mu_1 - \mu_2 = 0$ does not lie in conf. interval, reject

$H_0: \mu_1 \geq \mu_2 \quad \mu_2 \neq \leq \mu_1$

$H_1: \mu_1 < \mu_2 \quad \mu_2 > \mu_1$

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Q7: $n_1 = 10 \quad \bar{x}_1 = 290 \quad s_1 = 12 \quad \alpha = 0.05$

$n_2 = 16 \quad \bar{x}_2 = 321 \quad s_2 = 22 \quad \sigma_1^2 \neq \sigma_2^2$

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{(290-321)-(0)}{\sqrt{\frac{12^2}{10} + \frac{22^2}{16}}} = -4.64$$

$$d.f = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}} = \frac{\left(\frac{12^2}{10} + \frac{22^2}{16}\right)^2}{\frac{(12^2/10)^2}{10-1} + \frac{(22^2/16)^2}{16-1}} = 23.72 \approx 23$$

$P = \phi(|t_0|)$

$P = \phi(4.64) = 0.0005$

$P < \alpha$ so reject H_0.

Q8: $H_0: \sigma_1^2 = \sigma_2^2$

$H_1: \sigma_1^2 \neq \sigma_2^2$

$n_1 = 15 \quad n_2 = 15 \quad \frac{\alpha}{2} = \frac{0.05}{2} = 0.025$

$s_1^2 = 2.3 \quad s_2^2 = 1.9 \quad 1 - \frac{\alpha}{2} = 0.975 \approx 0.10$

$$F = \frac{s_1^2}{s_2^2} = \frac{2.3}{1.9} = 1.21 \qquad d.f = (14, 14)$$

$$\frac{s_1^2}{s_2^2} F_{0.10,14,14} \leq \frac{\sigma_1^2}{\sigma_2^2} \leq \frac{s_1^2}{s_2^2} F_{0.025,14,14}$$

$1/2.45$

$(1.21)(\quad) \leq \frac{\sigma_1^2}{\sigma_2^2} \leq (1.21)(3) \rightarrow \quad \leq \frac{\sigma_1^2}{\sigma_2^2} \leq 3.63$

2.45

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As $\frac{\sigma_1^2}{\sigma_2^2}$ lies in the confidence interval, do not reject H_0

Q9: $\hat{p}_1 = 0.78$ (14/18) $\quad \hat{p}_2 = 0.73$ (22/30) $\quad \hat{q}_1 = 0.22 \quad \hat{q}_2 = 0.27$

(a) $H_0: P_1 \leq P_2 \qquad \alpha = 0.05$

$H_1: P_1 > P_2$

$$z = \frac{(0.78 - 0.73) - (P_1 - P_2)}{\sqrt{\frac{(0.78)(0.22)}{18} + \frac{(0.73)(0.27)}{30}}} = 0.394$$

$P = 1 - \phi(z_0) = 1 - 0.651732 = 0.3483$

$P > \alpha$, do not reject H_0

(b) $\alpha = 0.05$

$$(\hat{p}_1 - \hat{p}_2) \; P_1 - P_2 > (\hat{p}_1 - \hat{p}_2) - z_{\alpha} \sqrt{\frac{\hat{p}_1 \hat{q}_1}{n_1} + \frac{\hat{p}_2 \hat{q}_2}{n_2}}$$

$$P1 - P2 > (0.78 - 0.73) - (1.65) \sqrt{\frac{(0.78)(0.22)}{18} + \frac{(0.73)(0.27)}{30}}$$

$P1 - P2 > 0.05 - 0.209$

$P1 - P2 > -0.159$

As $P1 - P2 = 0$ lies in the region, do not reject

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Q10: $H_0: \mu_1 = \mu_2 = \mu_3$ $n_1 = 8$ $n_2 = 7$ $n_3 = 6$

$H_1: \mu_1 \neq \mu_2 \neq \mu_3$ $T_1 = 1718$ $T_2 = 1362$ $T_3 = 1600$

Source of Variation	D.F	Sum of Squares	Mean Square	Test Stat
Between	2	SSB = 17642.02	MSB = 8821.01	F = 1.298
Within	18	SSW = 122344.548	MSW = 6796.92	
Total	20	SST = 139986.568		

$$SSB = \frac{T_1^2}{n_1} + \frac{T_2^2}{n_2} + \frac{T_3^2}{n_3} - \frac{(\Sigma x)^2}{n}$$

$$SSB = \frac{(1718)^2}{8} + \frac{(1362)^2}{7} + \frac{(1600)^2}{6} - \frac{(4680)^2}{21} = 17642.02$$

$$SSW = \Sigma x^2 - \left(\frac{T_1^2}{n_1} + \frac{T_2^2}{n_2} + \frac{T_3^2}{n_3} \right)$$

$$SSW = 1182958 - 1060613.452 = 122344.548$$

$$MSB = \frac{17642.02}{2} = 8821.01$$

$$MSW = \frac{122344.548}{18} = 6796.92$$

$$F = \frac{MSB}{MSW} = \frac{8821.01}{6796.92} = 1.298$$

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$F = 1.298 \quad \alpha = 0.05$

CR: $F \geq F_{\alpha, df}$

$F \geq F_{(0.05, (2, 18))}$

$F \geq 3.55$

As F is not in the critical region we cannot reject H_0.

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